

```
--- Working with NumPy Arrays ---
```

```
1D Array:
```

```
[1 2 3 4 5]
```

```
2D Array:
```

```
[[1 2 3]
```

```
 [4 5 6]]
```

```
Array addition (arr_1d + 5):
```

```
[ 6  7  8  9 10]
```

```
Array multiplication (arr_1d * 2):
```

```
[ 2  4  6  8 10]
```

```
Sum of 2D array elements: 21
```

```
Mean of 1D array elements: 3.0
```

```
Reshaped Array (5x1):
```

```
[[1]
```

```
 [2]
```

```
 [3]
```

```
 [4]
```

```
 [5]]
```

```
First element of 1D array: 1
```

```
First row of 2D array: [1 2 3]
```

--- Working with Pandas DataFrames ---

Initial DataFrame:

	Name	Age	City
0	Alice	25	New York
1	Bob	30	Paris
2	Charlie	35	London
3	David	40	Tokyo

Accessing 'Name' column:

0	Alice
1	Bob
2	Charlie
3	David

Name: Name, dtype: object

Accessing first row (using iloc):

Name	Alice
Age	25
City	New York

Name: 0, dtype: object

Filtered DataFrame (Age > 30):

	Name	Age	City
2	Charlie	35	London
3	David	40	Tokyo

DataFrame with Salary column:

	Name	Age	City	Salary
0	Alice	25	New York	50000
1	Bob	30	Paris	60000
2	Charlie	35	London	70000
3	David	40	Tokyo	80000

Descriptive statistics of the DataFrame:

	Age	Salary
count	4.000000	4.000000
mean	32.500000	65000.000000
std	6.454972	12909.944488
min	25.000000	50000.000000
25%	28.750000	57500.000000
50%	32.500000	65000.000000
75%	36.250000	72500.000000
max	40.000000	80000.000000

```
--- Reading Data from Various Sources ---
```

```
Data from CSV:
```

	id	value	category
0	1	10	A
1	2	20	B
2	3	30	C

```
Data from Excel:
```

	id	value	category
0	1	10	A
1	2	20	B
2	3	30	C

```
Reading data from web (URL): https://raw.githubusercontent.com/cs109/2014\_data/master/countries.csv
```

```
Data from Web (first 5 rows):
```

	Country	Region	Population	Area	Immigration	Life Expectancy	GDP
0	Afghanistan	ASIA	31889923	652090	0.0	43.828	974.580338
1	Albania	EUROPE	3600523	28748	1650.0	76.423	5937.029526
2	Algeria	AFRICA	33333216	2381741	0.0	72.301	6223.367465
3	Angola	AFRICA	12420476	1246700	0.0	42.731	4797.231267
4	Antigua and Barbuda	NORTH AMERICA	86754	442	0.0	76.384	19118.276510

```
[5 rows x 7 columns]
```

--- Descriptive Analytics using Iris Dataset ---

1. Dataset Info:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 150 entries, 0 to 149

Data columns (total 5 columns):

#	Column	Non-Null Count	Dtype
0	sepal_length	150 non-null	float64
1	sepal_width	150 non-null	float64
2	petal_length	150 non-null	float64
3	petal_width	150 non-null	float64
4	species	150 non-null	object

dtypes: float64(4), object(1)

memory usage: 6.0+ KB

None

2. Descriptive Statistics:

	sepal_length	sepal_width	petal_length	petal_width
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.054000	3.758667	1.198667
std	0.828066	0.433594	1.764420	0.763161
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

3. Species counts:

setosa 50

versicolor 50

virginica 50

Name: species, dtype: int64

4. Mean values grouped by species:

	sepal_length	sepal_width	petal_length	petal_width
species				
setosa	5.006	3.418	1.464	0.244
versicolor	5.936	2.770	4.260	1.326
virginica	6.588	2.974	5.552	2.026

5. Generating Pairplot... (Window may pop up)

Uncomment the plotting code in the script to view the pairplot visually.